

Exp No: 25 PRODUCER–CONSUMER PROBLEM USING SYSTEMC

BProgram:

```
// Code your testbench here.
// Uncomment the next line for SystemC modules.
// #include <systemc>
#include <systemc.h>
// Producer Module
SC_MODULE(Producer) {
    sc_fifo_out<int> out;
    void produce() {
        for (int i = 1; i <= 5; i++) {
            cout << "Produced: " << i << endl;
            out.write(i);
            wait(1, SC_SEC);
        }
    }
    SC_CTOR(Producer) {
        SC_THREAD(produce);
    }
};
// Consumer Module
SC_MODULE(Consumer) {
    sc_fifo_in<int> in;
    void consume() {
        int value;
        while (true) {
            in.read(value);
            cout << "Consumed: " << value << endl;
            wait(1, SC_SEC);
        }
    }
};
```

```
}  
}  
SC_CTOR(Consumer) {  
    SC_THREAD(consume);  
}  
};  
  
int sc_main(int argc, char* argv[]) {  
    sc_fifo<int> fifo(5);  
    Producer producer("Producer");  
    Consumer consumer("Consumer");  
    producer.out(fifo);  
    consumer.in(fifo);  
    sc_start(10, SC_SEC);  
    return 0;  
}
```

New

Run

Save

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testbench.cpp



```
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3 // #include <systemc>
4 #include <systemc.h>
5 // Producer Module
6 SC_MODULE(Producer) {
7     sc_fifo_out<int> out;
8     void produce() {
9         for (int i = 1; i <= 5; i++) {
10             cout << "Produced: " << i << endl;
11             out.write(i);
12             wait(1, SC_SEC);
13         }
14     }
15     SC_CTOR(Producer) {
16         SC_THREAD(produce);
17     }
18 };
19 // Consumer Module
20 SC_MODULE(Consumer) {
21     sc_fifo_in<int> in;
22     void consume() {
23         int value;
24         while (true) {
```

design.

1
2
3

Log

Share

Log

Share

SystemC 2.3.3-Accellera --- May 23 2020 14:33:56
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Produced: 1

Consumed: 1

Produced: 2

Consumed: 2

Produced: 3

Consumed: 3

Produced: 4

Consumed: 4

Produced: 5

Consumed: 5